

Markov Chains and Random Walks: A Probabilistic Journey

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Motivation: Why Study This?

Problem: How do we model systems that evolve randomly over time?

- Stock prices fluctuating
- Weather patterns
- The spread of a rumor or disease
- The position of a wandering animal
- A user's navigation through a website

Insight: Many of these systems share a powerful simplifying property:

The Future is Independent of the Past, Given the Present.

This is the essence of a **Markov Chain** and **Random Walks**. **Why Study Them?**

- **Analytically Tractable:** Use linear algebra to understand long-term behavior.
- **Widely Applicable:** PageRank, speech recognition, bioinformatics, finance.
- **Intuitive:** Easy to visualize and understand.

A Motivating Example: The Weather

Scenario: Model the weather in a city. Three states:

- Rain (R)
- Normal (N)
- Sunny (S)

Transition Probabilities (from historical data):

- From **Rain**: 60% Rain, 30% Normal, 10% Sunny
- From **Normal**: 30% Rain, 20% Normal, 50% Sunny
- From **Sunny**: 10% Rain, 30% Normal, 60% Sunny

Transition Graph:

$$\begin{array}{lll} R \xrightarrow{0.6} R, & R \xrightarrow{0.3} N, & R \xrightarrow{0.1} S \\ N \xrightarrow{0.3} R, & N \xrightarrow{0.2} N, & N \xrightarrow{0.5} S \\ S \xrightarrow{0.1} R, & S \xrightarrow{0.3} N, & S \xrightarrow{0.6} S \end{array}$$

Key Insight: The probability of tomorrow's weather depends **only** on today's weather, not on the weather from the day before yesterday.

Definition: The Markov Property

Formal Definition: A **Markov Chain** is a sequence of random variables $X_0, X_1, X_2, \dots$ taking values in a state space $\mathcal{S}$, satisfying the **Markov Property**:

Markov Property

$$\Pr[X_{n+1} = x_{n+1} \mid X_n = x_n, X_{n-1} = x_{n-1}, \dots, X_0 = x_0] = \Pr[X_{n+1} = x_{n+1} \mid X_n = x_n]$$

In Plain English:

The probability of moving to the next state depends *only* on the current state. The entire history is irrelevant.

Key Conditions:

- 1 **Finite State Space:** $\mathcal{S}$ is finite (e.g., {Rain, Normal, Sunny}, or $\{1, 2, \dots, N\}$).
- 2 **Time-Homogeneity:** Transition probabilities are constant over time.

The Engine: The Transition Matrix

All the transition probabilities can be neatly organized into a **Transition Matrix $\mathbf{P}$** .

$\mathbf{P}$ is an $N \times N$ matrix where the entry P_{ij} is the probability of going from state i to state j in one step:

$$P_{ij} = \Pr[X_{n+1} = j \mid X_n = i]$$

Crucial Property: $\mathbf{P}$ is a Stochastic Matrix

- 1 All entries are non-negative.
- 2 Every row sums to 1. (From a given state, you must go to *some* state.)

Example: Weather Chain

$$\mathbf{P} = \begin{bmatrix} 0.6 & 0.3 & 0.1 \\ 0.3 & 0.2 & 0.5 \\ 0.1 & 0.3 & 0.6 \end{bmatrix}$$

Interpretation:

Multi-Step Transitions and Matrix Powers

Question: What is the probability of it being Sunny two days from now, given it is Rainy today?

Answer: Matrix multiplication!

Let $\pi^{(n)}$ be the probability distribution over states at time n . If $\pi^{(0)}$ is our starting distribution:

$$\pi^{(n)} = \pi^{(0)} \mathbf{P}^n$$

Two-Step Transition

The probability of going from state i to state j in exactly two steps is:

$$P_{ij}^{(2)} = \sum_{k \in \mathcal{S}} P_{ik} P_{kj}$$

This is simply the (i, j) entry of $\mathbf{P}^2$.

Visual: Summing Over All Intermediate States

State $i \xrightarrow{P_{ik}}$ State $k \xrightarrow{P_{kj}}$ State j

Random Walks on Graphs

A **Random Walk on a Graph** is a special and very important type of Markov Chain.

Given a graph $G = (V, E)$, from your current vertex, pick one of its neighbors uniformly at random and move to it.

Transition Matrix for Random Walk

If you are at vertex u with degree $d(u)$, you move to each neighbor v with probability $1/d(u)$.

$$P(u, v) = \begin{cases} \frac{1}{d(u)} & \text{if } (u, v) \in E, \\ 0 & \text{otherwise.} \end{cases}$$

Matrix Form:

$$\mathbf{W} = \mathbf{A}\mathbf{D}^{-1}$$

where $\mathbf{A}$ is the adjacency matrix and $\mathbf{D}$ is the degree matrix (diagonal with $D_{ii} = d(i)$).

The Question of Convergence

We can start a random walk at a specific vertex, but as time goes on, where do we end up?

A **stationary distribution** π is a probability distribution that remains unchanged by the transition matrix:

$$\pi \mathbf{P} = \pi$$

Interpretation: If the chain reaches the stationary distribution, it stays there forever.

Computing the Distribution: Starting with $\pi^{(0)}$, the distribution after t steps is: $\pi^{(t)} = \pi^{(0)} \mathbf{P}^t$

The Fundamental Question

As $t \rightarrow \infty$, does $\pi^{(t)}$ converge to a unique stationary distribution?

- If so, the chain “forgets” its initial state and reaches a steady-state.
- This is crucial for PageRank and Markov Chain Monte Carlo (MCMC).

When Does a Markov Chain Converge?

The Fundamental Theorem of Markov Chains

A Markov Chain on a finite state space is **ergodic** (it converges to a unique stationary distribution) **if and only if** it satisfies two conditions:

1. Irreducibility:

- The chain must be able to get from any state to any other state in some number of steps.
- For a random walk on an undirected graph: the graph must be **connected**.

2. Aperiodicity:

- The chain should not be trapped in a deterministic cycle.
- **Problem:** Bipartite graphs. In a bipartite graph, you can only return to your starting state after an even number of steps.
- **Solution:** The **Lazy Random Walk**. With probability $1/2$ you stay in place, with probability $1/2$ you move to a neighbor.

Stationary Distribution of a Random Walk

What is the Stationary Distribution for a Random Walk on an Undirected Graph?

The Answer

For a connected, non-bipartite graph $G = (V, E)$, with $m = |E|$ edges, the stationary distribution for vertex v is proportional to its degree $d(v)$:

$$\pi_v = \frac{d(v)}{2m}$$

Proof: Check $\pi \mathbf{P} = \pi$

$$(\pi \mathbf{P})_v = \sum_{u \in N(v)} \pi_u \cdot \frac{1}{d(u)} = \sum_{u \in N(v)} \frac{d(u)}{2m} \cdot \frac{1}{d(u)} = \sum_{u \in N(v)} \frac{1}{2m} = \frac{d(v)}{2m} = \pi_v$$

What does this mean?

- The chain is most likely to be found at vertices with a high degree (highly connected nodes).

The Spectral Gap: How Fast Do We Converge?

We know the chain *converges*, but **how fast**? This is determined by the second-largest eigenvalue.

For a random walk on an undirected graph, define the normalized Laplacian:

$$\mathbf{S} = \mathbf{D}^{-1/2} \mathbf{A} \mathbf{D}^{-1/2}$$

Spectral Gap $\lambda = 1 - |\omega_2|$, where ω_2 is the second-largest eigenvalue of $\mathbf{S}$.

Interpretation:

- **Large spectral gap** (λ close to 1) $\Rightarrow$ fast convergence (“mixing”).
- **Small spectral gap** (λ close to 0) $\Rightarrow$ slow convergence.

Examples:

- **Complete Graph** K_n : Fast mixing. Large spectral gap.
- **Path Graph** P_n : Slow mixing. Spectral gap $\propto 1/n^2$.
- **“Dumbbell” Graph**: Very slow mixing. Two cliques connected by a single bridge.

Hitting and Cover Times

Hitting Time $h_{u,v}$

The expected number of steps to first reach state v , starting from state u .

Asymmetry: Even in undirected graphs, hitting times are not symmetric!

$$h_{u,v} \neq h_{v,u}$$

Example: The Lollipop Graph

- A clique connected to a path.
- Start in the **clique** (u) and hit the end of the **path** (v): $\Theta(n^3)$ time.
- Start at the end of the **path** (v) and hit the **clique** (u): $\Theta(n^2)$ time.
- Reason: The walk frequently gets lost in the clique when trying to leave it.

Commute Time $C_{u,v} = h_{u,v} + h_{v,u}$ and the formula: for an undirected graph, $C_{u,v} = 2m \cdot R_{u,v}$, where $R_{u,v}$ is the **effective resistance** between u and v in an electrical network.

This connects probability, graph theory, and physics!

Summary & Key Takeaways

Main Concepts:

- ➊ **Markov Chains:** Stochastic processes with the **memoryless property**. The future depends only on the present.
- ➋ **Transition Matrix P :** The engine of the chain. A stochastic matrix whose powers give the probability of a walk over any number of steps.
- ➌ **Random Walks on Graphs:** A concrete and intuitive model. The walker picks a random neighbor at each step.
- ➍ **Stationary Distribution:** The long-term probability distribution. For an undirected graph, $\pi_v = d(v)/2m$.
- ➎ **Convergence:** Requires irreducibility and aperiodicity.
- ➏ **Spectral Gap:** Governs the speed of convergence (“mixing time”).

Why This Matters:

- **Search:** Google’s PageRank.
- **Sampling:** Markov Chain Monte Carlo (MCMC) methods for Bayesian statistics.
- **Machine Learning:** Spectral clustering, diffusion maps, and

Textbooks:

- Mitzenmacher and Upfal. *Probability and Computing*.
- Lawler and Limic. *Random Walk: A Modern Introduction*.
- Levin, Peres, and Wilmer. *Markov Chains and Mixing Times*.

Lecture Notes:

- MIT OpenCourseWare: 18.S096 Topics in Mathematics with Applications in Finance.
- MIT 6.854: Advanced Algorithms lecture notes on Random Walks.